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Auburn Research: Student Symposium
Tuesday, April 9th, Auburn University Student Center

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Title: Strike (with) a pose: neural networks are easily fooled by strange poses of familiar objects
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Description: Deep neural networks (DNNs) are increasingly common components of computer vision systems. When handling "familiar" data, DNNs are capable of superhuman performance; however, inputs that are dissimilar to previously encountered examples (but that are still easily recognized by humans) can cause DNNs to make catastrophic mistakes. Here, we present a framework for discovering DNN failures that harnesses 3D computer graphics. Using our framework and a self-assembled dataset of 3D objects, we investigate the vulnerability of DNNs to "strange" poses of well-known objects. For objects that are readily recognized by DNNs in their canonical poses, DNNs incorrectly classify 97% of their pose space. Further, DNNs are highly sensitive to slight pose perturbations; for example, rotating a correctly classified object as little as 8 can often cause a DNN to misclassify. Lastly, 75% to 99% of adversarial poses transfer to DNNs with different architectures and/or trained with different datasets.